

A Clean Verification of an Unsmoothed Adaptive Riemannian ADMM

Corrected technical note

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Abstract

We repair and verify a proposed unsmoothed ADMM-type method for the nonsmooth composite manifold problem

$$\min_{x \in \mathcal{M}} f(x) + g(Ax).$$

The method introduces the split variable $y = Ax$, updates y by the exact proximal map of the original nonsmooth function g , takes one Riemannian gradient step in the x -block, and uses a damped adaptive multiplier update. Under the compact embedded-manifold, smooth- f , and Lipschitz-convex- g assumptions used in the motivating Riemannian ADMM literature, the algorithm returns an ε -KKT point of the original split problem in $O(\varepsilon^{-3})$ iterations. Thus, in that Lipschitz setting, the result is a solution of the unsmoothed Riemannian ADMM question. For arbitrary proper closed convex, extended-valued, or non-Lipschitz g , the proof gives only partial progress because it requires bounded proximal subgradients.

Status. The submitted draft was aimed at a real problem: Li, Ma, and Srivastava proposed a Moreau-envelope RADMM and left open how to design ADMM for the original nonsmooth problem without smoothing. The proof below fixes two serious technical gaps in the draft. First, all retraction estimates are local and are used only after proving that the generated tangent steps remain in a uniform local neighborhood. Second, the logarithmic damping constant in the descent proof is corrected. The result is close in spirit to the later ARADMM theorem of Deng, Jin, Hu, and Wang; this note is meant as a rigorous verification and cleanup, not as a novelty claim over that work.

1 Problem and KKT residual

Let $\mathcal{M} \subset \mathbb{R}^n$ be a compact embedded Riemannian submanifold and let $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. We consider

$$\min_{x \in \mathcal{M}} F(x) := f(x) + g(Ax), \tag{1}$$

where f is smooth and g is convex but possibly nonsmooth. Introduce the standard ADMM splitting

$$\min_{x \in \mathcal{M}, y \in \mathbb{R}^m} f(x) + g(y) \quad \text{subject to} \quad Ax - y = 0. \tag{2}$$

We use the sign convention

$$\mathcal{L}_\rho(x, y, \lambda) := f(x) + g(y) - \langle \lambda, Ax - y \rangle + \frac{\rho}{2} \|Ax - y\|^2, \quad \rho > 0. \tag{3}$$

With this convention, the exact KKT system for (2) is

$$\text{Proj}_{T_x \mathcal{M}}(\nabla f(x) - A^\top \lambda) = 0, \quad -\lambda \in \partial g(y), \quad Ax - y = 0. \tag{4}$$

At feasible points this is equivalent to the usual first-order stationarity condition for $f + g \circ A$ restricted to $\mathcal{M}$.

Definition 1 (ε -KKT point). A triple $(x, y, \lambda) \in \mathcal{M} \times \mathbb{R}^m \times \mathbb{R}^m$ is an ε -KKT point of (2) if

$$\left\| \text{Proj}_{T_x \mathcal{M}}(\nabla f(x) - A^\top \lambda) \right\| \leq \varepsilon, \quad (5)$$

$$\text{dist}(-\lambda, \partial g(y)) \leq \varepsilon, \quad (6)$$

$$\|Ax - y\| \leq \varepsilon. \quad (7)$$

2 Assumptions and local geometry

Assumption 1 (Analytic assumptions). The following conditions hold.

(A1) $\mathcal{M} \subset \mathbb{R}^n$ is a nonempty compact embedded submanifold. A smooth retraction $R_x : T_x \mathcal{M} \rightarrow \mathcal{M}$ is fixed.

(A2) f is continuously differentiable on a neighborhood of $\mathcal{M}$, and ∇f is Lipschitz on that neighborhood. Define

$$f_* := \min_{x \in \mathcal{M}} f(x), \quad G_f := \max_{x \in \mathcal{M}} \|\nabla f(x)\|.$$

(A3) $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is finite-valued, closed, convex, and ℓ_g -Lipschitz. Equivalently, every convex subgradient satisfies $\|s\| \leq \ell_g$ for all $s \in \partial g(y)$ and all $y \in \mathbb{R}^m$.

(A4) $\|A\|_{\text{op}} \leq \sigma_A$.

Remark 1 (The role of Lipschitzness). The proof uses the Lipschitz assumption only to bound the subgradient generated by the exact proximal y -update. The same proof works under the weaker a posteriori assumption that those particular proximal subgradients are uniformly bounded. It does not, by itself, cover arbitrary extended-valued convex regularizers such as indicator functions of convex sets, where normal cones can be unbounded.

The following lemma states the geometric estimates in the precise local form needed below.

Lemma 1 (Uniform local retraction estimates). *Fix $Y, \Lambda \geq 0$. There exist constants $\delta_R > 0$ and $a, L_R, C_R > 0$, depending only on $Y, \Lambda, f, A, \mathcal{M}$, and the retraction, such that for every $\rho \geq 1$, every $x \in \mathcal{M}$, every $\eta \in T_x \mathcal{M}$ with $\|\eta\| \leq \delta_R$, every $y \in \mathbb{R}^m$ with $\|y\| \leq Y$, and every $\lambda \in \mathbb{R}^m$ with $\|\lambda\| \leq \Lambda$, the function*

$$\phi_{\rho, y, \lambda}(z) := f(z) - \langle \lambda, Az - y \rangle + \frac{\rho}{2} \|Az - y\|^2, \quad z \in \mathcal{M},$$

satisfies

$$\|R_x(\eta) - x\| \leq a \|\eta\|, \quad (8)$$

$$\phi_{\rho, y, \lambda}(R_x(\eta)) \leq \phi_{\rho, y, \lambda}(x) + \langle \text{grad } \phi_{\rho, y, \lambda}(x), \eta \rangle + \frac{L_R \rho}{2} \|\eta\|^2, \quad (9)$$

$$\|\text{grad } \phi_{\rho, y, \lambda}(R_x(\eta)) - \text{grad } \phi_{\rho, y, \lambda}(x)\| \leq C_R \rho \|\eta\|. \quad (10)$$

The two Riemannian gradients in (10) are viewed as ambient vectors in $\mathbb{R}^n$.

Proof. Because $\mathcal{M}$ is compact and R is smooth, after reducing δ_R if necessary the points $R_x(\eta)$ remain in a fixed compact neighborhood of $\mathcal{M}$, and $\|R_x(\eta) - x\| \leq a \|\eta\|$ uniformly. On that compact neighborhood,

$$\nabla_z \phi_{\rho,y,\lambda}(z) = \nabla f(z) - A^\top \lambda + \rho A^\top (Az - y)$$

has norm at most $c_0 \rho$ and ambient Lipschitz constant at most $c_1 \rho$, uniformly over $\rho \geq 1$, $\|y\| \leq Y$, and $\|\lambda\| \leq \Lambda$. The pullbacks $\eta \mapsto \phi_{\rho,y,\lambda}(R_x(\eta))$ therefore have gradient Lipschitz constant at most $L_R \rho$ on the local tangent ball, which gives (9). Finally, $\text{grad } \phi_{\rho,y,\lambda}(z) = P_z \nabla_z \phi_{\rho,y,\lambda}(z)$, where P_z is the orthogonal projector onto $T_z \mathcal{M}$. The map $z \mapsto P_z$ is Lipschitz on compact embedded submanifolds, and combining this with the ambient $O(\rho)$ bounds gives (10). $\square$

3 Algorithm

Use the proximal convention

$$\text{prox}_{g/\rho}(z) := \arg \min_y \left\{ g(y) + \frac{\rho}{2} \|y - z\|^2 \right\}.$$

Fix constants

$$c_\rho \geq 1, \quad c_\gamma > 0, \quad \Gamma > 0, \quad c_\tau > 0, \quad (11)$$

and define

$$\rho_k := c_\rho (k+1)^{1/3}, \quad \tau_k := \frac{c_\tau}{\rho_k}. \quad (12)$$

Start from $x^0 \in \mathcal{M}$, $y^0 \in \mathbb{R}^m$, and $\lambda^0 \in \mathbb{R}^m$. Given (x^k, y^k, λ^k) , perform the following updates.

Exact nonsmooth y -step.

$$y^{k+1} := \arg \min_{y \in \mathbb{R}^m} \mathcal{L}_{\rho_k}(x^k, y, \lambda^k) = \text{prox}_{g/\rho_k} \left(Ax^k - \frac{\lambda^k}{\rho_k} \right). \quad (13)$$

One Riemannian gradient step in x . Set

$$v^k := \text{grad}_x \mathcal{L}_{\rho_k}(x^k, y^{k+1}, \lambda^k) = \text{Proj}_{T_{x^k} \mathcal{M}} \left(\nabla f(x^k) - A^\top \lambda^k + \rho_k A^\top (Ax^k - y^{k+1}) \right), \quad (14)$$

and update

$$x^{k+1} := R_{x^k}(-\tau_k v^k). \quad (15)$$

Damped adaptive multiplier step. Let

$$r^{k+1} := Ax^{k+1} - y^{k+1}. \quad (16)$$

With the convention that $\Gamma/(0 \cdot (k+2)^2) = +\infty$, set

$$\gamma_{k+1} := \min \left\{ \frac{\Gamma}{\|r^{k+1}\| (k+2)^2}, \frac{c_\gamma}{\rho_k \log^2(k+2)} \right\}, \quad (17)$$

and update

$$\lambda^{k+1} := \lambda^k - \gamma_{k+1} r^{k+1}. \quad (18)$$

The method is unsmoothed: the only operation involving g is the exact proximal update (13) for the original g .

4 A priori bounds

Define

$$\Lambda := \|\lambda^0\| + \Gamma \sum_{j=0}^{\infty} \frac{1}{(j+2)^2}, \quad B := \ell_g + \Lambda, \quad A_M := \max_{x \in \mathcal{M}} \|Ax\|, \quad (19)$$

$$Y := A_M + B, \quad V := G_f + \sigma_A(\Lambda + B). \quad (20)$$

Apply Lemma 1 with this Y and Λ , and let δ_R, a, L_R, C_R be the resulting constants. Also set

$$d := a\sigma_A, \quad D_\rho := \frac{c_\rho^3}{3}, \quad q_0 := \frac{1}{\log^2 2}. \quad (21)$$

The concavity of $t^{1/3}$ gives

$$0 \leq \rho_{k+1} - \rho_k \leq \frac{D_\rho}{\rho_k^2}. \quad (22)$$

Lemma 2 (Multiplier, proximal residual, and step bounds). *Assume $c_\tau V \leq \delta_R$. Then, for all $k \geq 0$,*

$$\|\lambda^k\| \leq \Lambda, \quad (23)$$

$$\|Ax^k - y^{k+1}\| \leq \frac{B}{\rho_k}, \quad (24)$$

$$\|y^{k+1}\| \leq Y, \quad (25)$$

$$\|v^k\| \leq V, \quad (26)$$

$$\|r^{k+1}\| \leq d\tau_k \|v^k\| + \frac{B}{\rho_k}. \quad (27)$$

Moreover every generated retraction step satisfies $\tau_k \|v^k\| \leq \delta_R$.

Proof. By (17) and (18),

$$\|\lambda^{k+1} - \lambda^k\| = \gamma_{k+1} \|r^{k+1}\| \leq \frac{\Gamma}{(k+2)^2}.$$

Summing gives (23). The optimality condition of (13) is

$$0 \in \partial g(y^{k+1}) + \lambda^k - \rho_k(Ax^k - y^{k+1}). \quad (28)$$

Hence

$$s^{k+1} := \rho_k(Ax^k - y^{k+1}) - \lambda^k \in \partial g(y^{k+1}). \quad (29)$$

Since g is ℓ_g -Lipschitz, $\|s^{k+1}\| \leq \ell_g$. Thus

$$\rho_k \|Ax^k - y^{k+1}\| \leq \|s^{k+1}\| + \|\lambda^k\| \leq \ell_g + \Lambda = B,$$

which proves (24). Then (25) follows from $\|Ax^k\| \leq A_M$ and $\rho_k \geq 1$.

From (14), (23), and (24),

$$\|v^k\| \leq G_f + \sigma_A \Lambda + \rho_k \sigma_A \|Ax^k - y^{k+1}\| \leq G_f + \sigma_A(\Lambda + B) = V.$$

Therefore $\tau_k \|v^k\| \leq c_\tau V / \rho_k \leq c_\tau V \leq \delta_R$, so Lemma 1 applies to every generated retraction step. Finally,

$$r^{k+1} = A(x^{k+1} - x^k) + (Ax^k - y^{k+1}),$$

and (8) gives (27). $\square$

5 Descent

Define the summable error constant

$$E := 2B^2 \sum_{k=0}^{\infty} \left(\frac{c_\gamma}{\rho_k^3 \log^2(k+2)} + \frac{D_\rho}{2\rho_k^4} \right) < \infty. \quad (30)$$

The first series is comparable to $\sum_k 1/(k \log^2 k)$, and the second is comparable to $\sum_k k^{-4/3}$.

Lemma 3 (Almost descent). *Assume*

$$c_\tau V \leq \delta_R, \quad c_\tau \leq \min \left\{ \frac{1}{L_R}, \frac{1}{8d^2(q_0 c_\gamma + D_\rho/2)} \right\}, \quad (31)$$

where the last term is interpreted as $+\infty$ if $d = 0$. Then, for all $k \geq 0$,

$$\begin{aligned} & \mathcal{L}_{\rho_{k+1}}(x^{k+1}, y^{k+1}, \lambda^{k+1}) - \mathcal{L}_{\rho_k}(x^k, y^k, \lambda^k) \\ & \leq -\frac{\tau_k}{4} \|v^k\|^2 + 2B^2 \left(\frac{c_\gamma}{\rho_k^3 \log^2(k+2)} + \frac{D_\rho}{2\rho_k^4} \right). \end{aligned} \quad (32)$$

Proof. The y -step minimizes $\mathcal{L}_{\rho_k}(x^k, \cdot, \lambda^k)$, hence

$$\mathcal{L}_{\rho_k}(x^k, y^{k+1}, \lambda^k) \leq \mathcal{L}_{\rho_k}(x^k, y^k, \lambda^k). \quad (33)$$

Let $\phi_k(x) := \mathcal{L}_{\rho_k}(x, y^{k+1}, \lambda^k)$, ignoring the constant $g(y^{k+1})$ when applying Lemma 1. By Lemma 2, the step is in the local retraction neighborhood. Therefore (9), (15), and $c_\tau \leq 1/L_R$ give

$$\begin{aligned} \mathcal{L}_{\rho_k}(x^{k+1}, y^{k+1}, \lambda^k) & \leq \mathcal{L}_{\rho_k}(x^k, y^{k+1}, \lambda^k) - \tau_k \|v^k\|^2 + \frac{L_R \rho_k \tau_k^2}{2} \|v^k\|^2 \\ & \leq \mathcal{L}_{\rho_k}(x^k, y^{k+1}, \lambda^k) - \frac{\tau_k}{2} \|v^k\|^2. \end{aligned} \quad (34)$$

The multiplier update increases the Lagrangian by

$$\mathcal{L}_{\rho_k}(x^{k+1}, y^{k+1}, \lambda^{k+1}) - \mathcal{L}_{\rho_k}(x^{k+1}, y^{k+1}, \lambda^k) = \gamma_{k+1} \|r^{k+1}\|^2, \quad (35)$$

and the penalty increase contributes

$$\mathcal{L}_{\rho_{k+1}}(x^{k+1}, y^{k+1}, \lambda^{k+1}) - \mathcal{L}_{\rho_k}(x^{k+1}, y^{k+1}, \lambda^{k+1}) = \frac{\rho_{k+1} - \rho_k}{2} \|r^{k+1}\|^2. \quad (36)$$

Combining (33)–(36) gives

$$L_{k+1} - L_k \leq -\frac{\tau_k}{2} \|v^k\|^2 + \left(\gamma_{k+1} + \frac{\rho_{k+1} - \rho_k}{2} \right) \|r^{k+1}\|^2, \quad (37)$$

where $L_j := \mathcal{L}_{\rho_j}(x^j, y^j, \lambda^j)$.

Using (17), (22), (27), and $(u+w)^2 \leq 2u^2 + 2w^2$,

$$\begin{aligned} & \left(\gamma_{k+1} + \frac{\rho_{k+1} - \rho_k}{2} \right) \|r^{k+1}\|^2 \\ & \leq 2d^2 \tau_k^2 \left(\frac{c_\gamma}{\rho_k \log^2(k+2)} + \frac{D_\rho}{2\rho_k^2} \right) \|v^k\|^2 + 2B^2 \left(\frac{c_\gamma}{\rho_k^3 \log^2(k+2)} + \frac{D_\rho}{2\rho_k^4} \right). \end{aligned}$$

Since $1/\log^2(k+2) \leq q_0$, $\rho_k \geq 1$, and $\tau_k = c_\tau/\rho_k$, condition (31) implies

$$2d^2 \tau_k^2 \left(\frac{c_\gamma}{\rho_k \log^2(k+2)} + \frac{D_\rho}{2\rho_k^2} \right) \leq \frac{\tau_k}{4}.$$

Substitution into (37) proves (32). $\square$

6 Complexity theorem

Since g is finite-valued convex and Lipschitz, it is continuous. Define

$$g_Y := \min_{\|y\| \leq Y} g(y) > -\infty. \quad (38)$$

For every $k \geq 1$, Lemma 2 gives $\|y^k\| \leq Y$. Completing the square in (3) gives, for every $k \geq 1$,

$$\mathcal{L}_{\rho_k}(x^k, y^k, \lambda^k) = f(x^k) + g(y^k) + \frac{\rho_k}{2} \left\| Ax^k - y^k - \frac{\lambda^k}{\rho_k} \right\|^2 - \frac{\|\lambda^k\|^2}{2\rho_k} \geq L_{\min}, \quad (39)$$

where

$$L_{\min} := f_* + g_Y - \frac{\Lambda^2}{2\rho_0}. \quad (40)$$

Set

$$G := 4 \left(\mathcal{L}_{\rho_0}(x^0, y^0, \lambda^0) - L_{\min} + E \right). \quad (41)$$

Theorem 1 (Unsmoothed adaptive RADMM complexity). *Suppose Assumption 1 holds and choose c_τ so that*

$$c_\tau V \leq \delta_R, \quad c_\tau \leq \min \left\{ \frac{1}{L_R}, \frac{1}{C_R}, \frac{1}{8d^2(q_0 c_\gamma + D_\rho/2)} \right\}, \quad (42)$$

where the last term is interpreted as $+\infty$ if $d = 0$. Let $K \geq 2$ and choose

$$\kappa \in \{ \lceil K/2 \rceil, \dots, K-1 \}$$

satisfying

$$\|v^\kappa\|^2 \leq \frac{\sum_{j=\lceil K/2 \rceil}^{K-1} \tau_j \|v^j\|^2}{\sum_{j=\lceil K/2 \rceil}^{K-1} \tau_j}. \quad (43)$$

Define

$$\hat{x} := x^{\kappa+1}, \quad \hat{y} := y^{\kappa+1}, \quad \hat{\lambda} := \lambda^\kappa - \rho_\kappa(Ax^{\kappa+1} - y^{\kappa+1}). \quad (44)$$

Let

$$C_v := 2\sqrt{\frac{c_\rho G}{c_\tau}}. \quad (45)$$

Then

$$\left\| \text{Proj}_{T_{\hat{x}}\mathcal{M}}(\nabla f(\hat{x}) - A^\top \hat{\lambda}) \right\| \leq 2C_v(K+1)^{-1/3}, \quad (46)$$

$$\text{dist}(-\hat{\lambda}, \partial g(\hat{y})) \leq dc_\tau C_v(K+1)^{-1/3}, \quad (47)$$

$$\|A\hat{x} - \hat{y}\| \leq \left(dc_\tau C_v + \frac{2^{1/3}B}{c_\rho} \right) (K+1)^{-1/3}. \quad (48)$$

Consequently, the method returns an ε -KKT point in $O(\varepsilon^{-3})$ iterations.

Proof. Summing Lemma 3 from $k = 0$ to N and using (30) and (39) at $N+1$ gives

$$\sum_{k=0}^N \tau_k \left\| v^k \right\|^2 \leq G, \quad N \geq 0. \quad (49)$$

The index set $\{\lceil K/2 \rceil, \dots, K-1\}$ has at least $(K+1)/4$ indices for $K \geq 2$. Also, for every $j \leq K-1$, $\rho_j \leq c_\rho(K+1)^{1/3}$. Therefore

$$\sum_{j=\lceil K/2 \rceil}^{K-1} \tau_j \geq \frac{c_\tau}{4c_\rho}(K+1)^{2/3}. \quad (50)$$

Equations (43), (49), and (50) imply

$$\|v^\kappa\| \leq C_v(K+1)^{-1/3}. \quad (51)$$

Let $\phi_\kappa(x) := \mathcal{L}_{\rho_\kappa}(x, y^{\kappa+1}, \lambda^\kappa)$, again ignoring the constant $g(y^{\kappa+1})$ in the x -gradient. Then $v^\kappa = \text{grad } \phi_\kappa(x^\kappa)$ and $\hat{x} = R_{x^\kappa}(-\tau_\kappa v^\kappa)$. By (10) and $c_\tau \leq 1/C_R$,

$$\|\text{grad } \phi_\kappa(\hat{x})\| \leq \|v^\kappa\| + C_R \rho_\kappa \tau_\kappa \|v^\kappa\| \leq 2\|v^\kappa\|.$$

Moreover, by the definition of $\hat{\lambda}$,

$$\text{grad } \phi_\kappa(\hat{x}) = \text{Proj}_{T_{\hat{x}}\mathcal{M}} \left(\nabla f(\hat{x}) - A^\top \hat{\lambda} \right).$$

Together with (51), this proves (46).

The y -optimality condition (28) at iteration κ gives

$$-\tilde{\lambda} \in \partial g(\hat{y}), \quad \tilde{\lambda} := \lambda^\kappa - \rho_\kappa(Ax^\kappa - \hat{y}). \quad (52)$$

Since $\hat{\lambda} - \tilde{\lambda} = -\rho_\kappa A(\hat{x} - x^\kappa)$,

$$\begin{aligned} \text{dist}(-\hat{\lambda}, \partial g(\hat{y})) &\leq \|\hat{\lambda} - \tilde{\lambda}\| \leq \rho_\kappa \sigma_A \|\hat{x} - x^\kappa\| \\ &\leq \rho_\kappa \sigma_A a \tau_\kappa \|v^\kappa\| = d c_\tau \|v^\kappa\|. \end{aligned}$$

Using (51) proves (47).

Finally, Lemma 2 gives

$$\|A\hat{x} - \hat{y}\| \leq d\tau_\kappa \|v^\kappa\| + \frac{B}{\rho_\kappa}.$$

Since $\tau_\kappa \leq c_\tau$ and $\kappa \geq \lceil K/2 \rceil$ implies $1/\rho_\kappa \leq 2^{1/3}/[c_\rho(K+1)^{1/3}]$, (51) proves (48). Choosing K proportional to ε^{-3} makes all three residuals at most ε . $\square$

7 Scope and partial-progress statement

The theorem proves convergence to an approximate KKT point for the original unsmoothed split problem (2). It does not introduce a Moreau envelope objective, a smoothing homotopy, or a differentiable surrogate for g . The only nonsmooth operation is the exact proximal minimization of the original g in (13). Under Assumption 1, this is a direct unsmoothed ADMM-type solution with $O(\varepsilon^{-3})$ iteration complexity.

The theorem should not be advertised as a solution for arbitrary proper closed convex g . The proof extends to such functions only if the proximal subgradients

$$s^{k+1} := \rho_k(Ax^k - y^{k+1}) - \lambda^k \in \partial g(y^{k+1})$$

are known a priori to be uniformly bounded. Without such a bound, the estimate $\|Ax^k - y^{k+1}\| = O(1/\rho_k)$ can fail, and the descent proof no longer controls feasibility or the augmented-Lagrangian ascent terms.

Proposition 1 (A posteriori extension). *Assume all hypotheses above except finite-valued Lipschitzness of g , and instead let $g : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, closed, convex, and lower bounded. If the iterates are well-defined and the proximal subgradients in the display above satisfy $\sup_k \|s^{k+1}\| < \infty$, then Theorem 1 remains valid with ℓ_g replaced by that finite supremum.*

Proof. The only use of Lipschitzness of g in the proof is the bound $\|s^{k+1}\| \leq \ell_g$ in Lemma 2. Replacing it by a uniform bound S gives the same proof with $B = S + \Lambda$. $\square$

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